

ANALYTICAL CHEMISTRY
(SPECIAL THEORY 1)

2018

SHORT QUESTIONS

- Q1-**What is the effect of concentration of ionic size on conduction?
Q2- Distinguish between automatic devices and automated devices?
Q3- Distinguish between limiting current and diffusion current?
Q4- Give any two advantages of amperometry?
Q5- What do you mean by residual current?
Q6- What are the capillary characteristics and give factors affecting them?
Q7- Give difference between continuous analyzer and discrete analyzer?
Q8- What are the advantages of conductometric titrations?
Q9- What is the role of supporting electrolyte in voltammetry?
Q10- What do you know about electric arc?

LONG QUESTIONS

- Q1-** Discuss the analytical applications of conductometry?
Q2- What is diffusion current in polarography? Discuss factors affecting it?
Q3- Explain different types of titration curves obtained in amperometry?
Q4- Write a note on electric arc?
Q5- Discuss the influence of Kinetics of Electron Transfer and mass transport on Faradaic current?
Q6- Discuss amperometric biosensors?

ANALYTICAL CHEMISTRY
(SPECIAL THEORY 1)

2016

SHORT QUESTIONS

- Q1-** Describe the conductometric titration of a mixture of strong and weak acid with a strong base?
- Q2-** What are the advantages of Amperometric titrations?
- Q3-** How specific conductivity of solution is determined?
- Q4-** What do you know about Faradaic Current in voltametry?
- Q5-** Give differences between specific conductance and molar conductance?
- Q6-** What is residual current? Give its two components?
- Q7-** What is the role of supporting electrolyte in voltametry?
- Q8-** What is electric arc? Give its uses?
- Q9-** What do you know about electric arc?
- Q10-** What are amperometric biosensors?

LONG QUESTIONS

- Q1-** Explain anodic stripping voltametry?
- Q2-** Describe various kinds of currents contributing to polarographic waves?
- Q3-** Describe the functions of three types of electrodes in classical polarography?
- Q4-** Discuss amperometric titrations along with their advantages?
- Q5-** Discuss various factors affecting conductance of solution?
- Q6-** Write a note on electric spark?

ANALYTICAL CHEMISTRY
(SPECIAL THEORY 1)

2015

SHORT QUESTIONS

- Q1-** Give the types of conductometric titrations?
- Q2-** What are the advantages of electroanalytical techniques over other analytical methods?
- Q3-** What change in conductance is observed when NH_4OH is added to HCl ?
- Q4-** How voltametric techniques are classified?
- Q5-** Give differences between equivalent conductance and molar conductance?
- Q6-** What is the difference between cathodic and anodic current?
- Q7-** What is the role of supporting electrolyte in voltametry?
- Q8-** What is electric arc? Give its uses?
- Q9-** How does electric spark occur?
- Q10-** What are the advantages of polarography?

LONG QUESTIONS

- Q1-** Discuss cyclic voltametry?
- Q2-** Describe various kinds of currents contributing to polarographic waves?
- Q3-** Give advantages and limitations of using DME?
- Q4-** Give applications of amperometric titrations?
- Q5-** Discuss electrodes in Amperometry?
- Q6-** Discuss applications of polarography in inorganic analysis?

ANALYTICAL CHEMISTRY
(SPECIAL THEORY 1)

2014

SHORT QUESTIONS

- Q1-** Define specific and molar conductance?
Q2- Write names of some methods used to determine the concentration of unknown solution in classical polarography?
Q3- What are the advantages of anodic stripping voltametry?
Q4- What is diffusion current? Write mathematical form of Fick's second law?
Q5- Draw three types of amperometric titrations?
Q6- Describe two methods for measurement of electrolytic conductance?
Q7- Differentiate between the arc and spark?
Q8- Why mercury electrode is widely used in voltametry?
Q9- Explain the importance of supporting electrolyte?
Q10- Differentiate between automated and automatic instruments?

LONG QUESTIONS

- Q1-** Describe the working and advantages of dropping mercury electrode?
Q2- Describe the factors affecting the conductance of solution?
Q3- Write down the advantages of automation process?
Q4- Write about the instrumentation for voltametry?
Q5- Explain the characteristics of a polarogram?